

Key points

- Maintenance immunosuppression is intended to prevent acute rejection and loss of the allograft while minimizing the risk of infection, malignancy, and drug side effects.
- Maintenance immunosuppressant regimens are combinations of two to four drugs with different mechanisms of action.
- Drugs available for maintenance immunosuppression are calcineurin inhibitors, antimetabolites, glucocorticoids, costimulation blocking agents, and signal proliferation inhibitors.
- The initial maintenance regimen for most patients consists of a calcineurin inhibitor, an antimetabolite, and a glucocorticoid – most frequently tacrolimus, mycophenolate mofetil, and prednisone.
- Modification of the initial regimen may be required due to development of complications, such as toxicity, allograft rejection, infection, malignancy, or pregnancy.
- The drugs have narrow therapeutic windows. Therapeutic drug monitoring for optimizing therapy (optimal levels for efficacy and minimizing toxicity).

Key points

- **The calcineurin inhibitors (CNIs), cyclosporine and tacrolimus**, form a ternary immunophilin-CNI-calcineurin complex within the cytosol, which prevents dephosphorylation and activation of cytosolic NFAT, a transcription factor critical in regulating gene transcription and production of cytokines and T cell receptor-mediated signaling.
 - Calcineurin is a calmodulin-dependent serine/threonine protein phosphatase.
 - Immunophilins are cytosolic isomerases.
 - Cyclosporine binds to cyclophilin. Tacrolimus binds FK binding protein (FKBP).
- CNI side effects include nephrotoxicity, hypertension, hyperlipidemia, hyperglycemia, hyperkalemia, and neurotoxicity.
- CNIs are taken orally once daily (IV formulations are available). They are substrates of CYP3A4 and P-glycoprotein. There is a high potential for drug interactions.
- Tacrolimus is generally preferred. It is associated with lower acute rejection rates and is better tolerated than cyclosporine.

Key points

- **The antimetabolites, mycophenolate mofetil (MMF) and azathioprine (AZA)**, interfere with de novo purine synthesis resulting in inhibition of T cell and B cell proliferation.
- **Mycophenolate** is a potent reversible noncompetitive inhibitor of inosine monophosphate dehydrogenase which blocks the synthesis of guanine monophosphate.
- The most common side effects of MMF include bone marrow suppression, mainly neutropenia, nausea, abdominal cramping, and persistent diarrhea. *Mycophenolate is teratogenic*. MMF is an oral drug (IV available) twice daily. It is hydrolyzed in the liver to active mycophenolic acid, which is excreted in urine as glucuronide metabolites.
- MMF is preferred because of superior efficacy and better side effect profile.
- **AZA** is converted to 6-mercaptopurine and then to the thioinosine monophosphate and 6-thioguanine toxic nucleotides which are incorporated during DNA synthesis.
- The most common AZA side effect is bone marrow suppression manifested by leukopenia. Other side effects, which are seen mainly at higher doses, include nausea, diarrhea, and hepatotoxicity (very high doses). AZA is an oral drug, once daily dosing. 6-MP is inactivated by xanthine oxidase and thiopurine methyltransferase.

Key points

- **Belatacept** is a selective T cell costimulation blocker. It is a recombinant fusion protein of CTLA-4 that binds CD80/86 on antigen presenting cells. It induces immunosuppression by preventing the costimulatory T cell activating signal – binding of CD28 on T cells to CD80/86.
- Belatacept is an alternative to a CNI for prevention of acute allograft rejection in EBV seropositive patients with biopsy-proven CNIs nephrotoxicity. It is given IV every 4 weeks (maintenance phase) in combination with oral MMF and a glucocorticoid.
- Belatacept use is associated with increased risk of EBV-associated post-transplant lymphoproliferative disorder (PTLD); risk is higher in EBV seronegative patients. Common side effects include edema, hypertension, headache, diarrhea, constipation, anemia, and cough.
- Belatacept is **contraindicated** in EBV seronegative patients. Use for liver transplant patients is not recommended due to an increased risk of graft loss and death.
- Information on the use of belatacept in pregnancy is limited.

Key points

- **mTOR inhibitors, sirolimus and everolimus**, are proliferation signal inhibitors. They bind to FK-binding protein and block the activity of mTOR, a cytoplasmic protein kinase, which interrupts DNA and protein synthesis, which prevents progression from the G1 to S phase causing cell cycle arrest, which prevents proliferation of T cells.
- mTOR inhibitors are generally used as alternatives in patients who cannot tolerate CNIs due to toxicity.
- Adverse effects include bone marrow suppression (anemia, leukopenia, thrombocytopenia), dyslipidemia, poor wound healing (antiangiogenic effect), hepatotoxicity, pneumonitis, and increased post-transplant mortality, in addition to nausea, vomiting, and diarrhea or constipation.
- Sirolimus use is not recommended in liver or lung transplantation because safety and efficacy have not been established.
- They are taken orally twice daily. They are substrates of CYP3A4 and P-glycoprotein.
- mTOR inhibitors are contraindicated in pregnancy.

Cyclosporine / Tacrolimus Uses

- Highly efficacious immunosuppressive drugs in routine use
- Prophylaxis of solid organ allograft rejection
- Treatment of GVHD after hematopoietic stem cell transplantation
- Additional uses:
 - Some autoimmune or inflammatory diseases
 - Cyclosporine ophthalmic drops for uveitis
 - Tacrolimus topically for atopic dermatitis, psoriasis

Tacrolimus is preferred by many transplant centers.

Compared with cyclosporine, tacrolimus is associated with decreased acute rejection rates, does not lower mycophenolate levels, is better tolerated by patients, and overall costs are similar.

CNI Toxicities: Cyclosporine / Tacrolimus

Class Toxicities Manifestations

Nephrotoxicity	• Vasoconstrictor effect on afferent renal arteriole • Interstitial fibrosis of renal parenchyma Nephrotoxicity is common in all transplant patients, both renal and nonrenal.	
Hypertension	• Renal vasoconstriction • Sodium retention	
Neurotoxicity	Tremor, headache, insomnia are common. (Other symptoms are less frequent.)	
Hyperkalemia	• Tubular insensitivity to aldosterone • Reduced excretion (K⁺ channel block)	
Hepatotoxicity	• Hyperbilirubinemia • Elevated serum transaminases	
Endocrine	Hyperglycemia (tacrolimus greater risk)	Cyclosporine: • Gingival hyperplasia • Hirsutism
GI	Diarrhea, nausea, vomiting	
Immune system	Infection / malignancy	

Calcineurin Inhibitors: Drug Interactions

High potential

Metabolic and transport related:

- CYP3A4 / P-glycoprotein inducers or inhibitors
- Cyclosporine reduces mycophenolate serum concentrations

Likely mechanism:

Cyclosporine inhibits MRP2 efflux transporter which inhibits biliary excretion of active mycophenolic acid (MPA), thus reduces MPA enterohepatic circulation.

Toxicity related:

Drugs that cause additive toxicity

- Nephrotoxicity
- Neurotoxicity
- Hyperkalemia
- and so on

Note: Cyclosporine and Tacrolimus are not used together – increased risk of toxicities.

Mycophenolate mofetil pharmacokinetics and therapeutic uses

Pharmacokinetics:

Mycophenolate mofetil (MMF) is a prodrug of mycophenolic acid (MPA).

- Oral, IV
- MMF is rapidly and completely converted in the liver to active mycophenolic acid (MPA)
- MPA bioavailability 94%
- Enterohepatic circulation of MPA
- Hepatic glucuronidation of MPA → MPA glucuronide (MPAG)
- MPAG excreted in urine
- MPA $t_{1/2}$ mean ~18 hrs

Therapeutic uses:

- Prophylaxis of organ rejection

used in 3-drug combination:

CNI+ MMF + GC

MMF has superior efficacy and better side effect profile than AZA and is the preferred drug for transplantation rejection prophylaxis.

- Prophylaxis and treatment GVHD
- Treatment of psoriasis, lupus, myasthenia gravis, other autoimmune diseases

MPA Sodium enteric coated tablet has different absorption characteristics and may produce less gastric upset.

Mycophenolate Toxicities

Most common:

- Bone marrow suppression
Anemia and leukopenia most frequent
- Persistent diarrhea, nausea / vomiting
(dose-related)

Risk of serious toxicities is typical of immunosuppressants class including infection and malignancy.

- Pregnancy: Avoid

Increased risk of first trimester pregnancy loss and congenital malformations.

Drug interactions

Avoid or use with caution (monitor).

Immunosuppressants

Enhanced immune suppression increases risk of infection and malignancy

Myelosuppressants

Enhanced bone marrow suppression increases risk of leukopenia, anemia, thrombocytopenia

Cyclosporine

Reduces serum levels of mycophenolic acid.

Cholestyramine Bile acid sequestrants

Chelates drug in gut and reduces absorption

Aluminum / magnesium hydroxide

Antacids reduce absorption. Separate by two hours.

Azathioprine

Pharmacokinetics

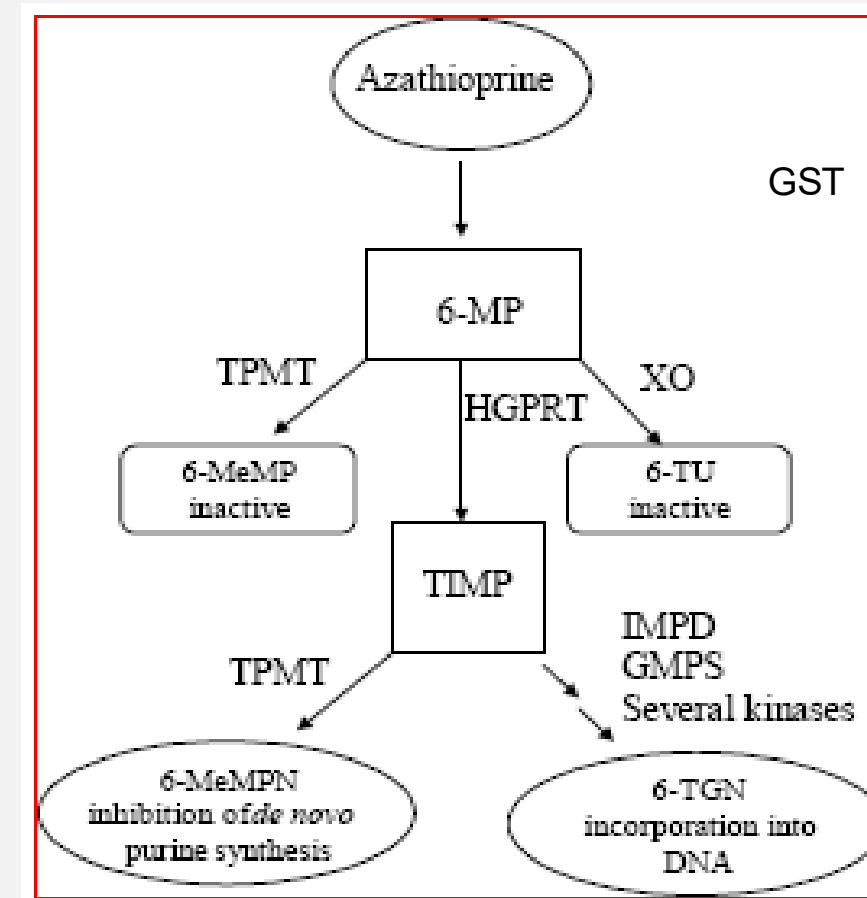
- Oral PRODRUG:
AZA glutathione-S-transferase → 6-MP
- Metabolism to inactive metabolites by
Xanthine oxidase (XO) and
Thiopurine methyltransferase (TPMT)
- Excretion of metabolites in urine
- $t_{1/2}$ of 6-MP ~1 hour

Pharmacogenomics

- Patients deficient in TPMT → at risk for
severe life-threatening myelosuppression

Therapeutic uses

- Prophylaxis of acute allograft rejection
calcineurin inhibitor + AZA + glucocorticoid
- Inflammatory and autoimmune diseases



6-TIMP → 6-TGMP →→ 6-TGTP
↓
6-TGTP is incorporated into DNA of proliferating cells
↓
induces strand breaks and base mispairing
↓
cell death

Azathioprine Toxicities / Drug Interactions

Most serious:

- Bone marrow suppression with profound leukopenia
- Hepatotoxicity with high doses especially in pts with preexisting hepatic dysfunction
- GI effects
- Other potential for toxicities typical of the immunosuppressants class.
- Pregnancy: AZA has been used safely

Allopurinol | Febuxostat
Xanthine oxidase inhibitors

↓ AZA inactivation

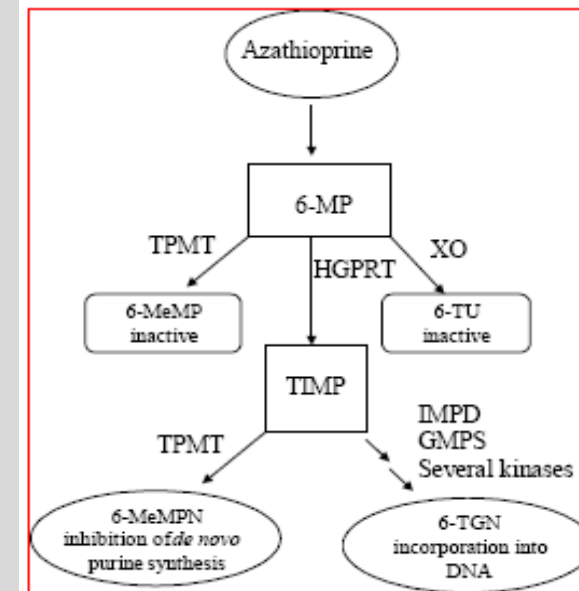


shunt AZA to HGPRT pathway



increase AZA toxicity risk

Remember increased toxicity risk in patients with TPMT deficient activity.



Belatacept

Therapeutic uses

- Maintenance therapy for prophylaxis of acute organ rejection in kidney transplant patients who are seropositive for EBV
- Combination with basiliximab, mycophenolate, and corticosteroids

Adverse effects:

- Post-transplant lymphoproliferative disorder (PTLD), primarily involving the CNS
- Reactivation of latent viral infections: Progressive multifocal leukoencephalopathy (PML) by activation of JC virus, and other serious infections
- Other adverse effects typical of the immunosuppressants class and biologic drugs.

Caution / Contraindication:

Belatacept should not be given to EBV-seronegative patient or unknown EBV status. The risk of PTLD is higher for patients who have not been exposed to EBV.

Sirolimus Therapeutic Uses

Transplant uses:

- Alternative for kidney transplant rejection prophylaxis
- Various (off-label) organ transplantation protocols in patients with CNI-related nephrotoxicity

Safety and efficacy in liver transplantation and lung transplantation have not been established. Use not recommended.

- Acute and chronic graft v. host disease (GVHD) treatment and prevention (off-label)

Non-transplantation uses of sirolimus:

- Drug-eluting coronary stents: Shown to reduce restenosis and additional adverse cardiac events in patients with severe coronary artery disease, due to its antiproliferative effects (prevents smooth muscle proliferation)
- Lymphangioleiomyomatosis: Lung disease caused by abnormal growth of smooth muscle cells in the lungs and lymphatics
- Topical use: Facial angiofibroma associated with tuberous sclerosis

Sirolimus toxicities and Drug Interactions

Adverse effects:

- Profound myelosuppression: Thrombocytopenia (common); anemia; leukopenia
- Impaired wound healing, wound dehiscence
 - May decrease angiogenesis growth factors and fibroblast proliferation.
- Hyperlipidemia: Hypertriglyceridemia and hypercholesterolemia are common.
- Peripheral edema, lymphedema, pleural effusion, pericardial effusions
- Interstitial lung disease (pneumonitis, bronchiolitis obliterans)
- Diarrhea, nausea, vomiting; headache; arthralgia; aphthous ulcers; acne
- Infection risk increased
- Hypersensitivity reactions

Cyclosporine or Tacrolimus + Sirolimus: Avoid combination

- ↑ CNI nephrotoxicity with deterioration of renal function
- ↑ risk of adverse drug events, including mortality
- Cyclosporine may ↑ risk of CNI-induced hemolytic uremic syndrome
- Sirolimus decreases tacrolimus blood levels and AUC (30% decrease)
- CYP3A4 / P-gp inducers / inhibitors

- Maintenance immunosuppressive agents are intended to provide adequate immunosuppression to prevent allograft rejection and minimize infection, malignancy, and drug-related adverse effects. Regimens typically include 2 to 4 drugs.
- The main classes are the calcineurin inhibitors, antiproliferative antimetabolites, costimulation blockers, and proliferation signaling inhibitors (mTOR inhibitors).
- The calcineurin inhibitors are cyclosporine and tacrolimus, which block signal transduction through the calcium-calcineurin pathway, thus impairing T cell activation. Calcineurin is a calcium-dependent protein phosphatase that dephosphorylates NFAT, which enters the T cell nucleus and induces transcription and production of cytokines, including IL-2.
- Cyclosporine binds cyclophilin, while tacrolimus binds FKBP-12. The drug-immunophilin complexes inhibit the activity of calcineurin. Tacrolimus is preferred by many transplant centers due to its decreased rate of allograft rejection.

- Cyclosporine and tacrolimus are substrates of the CYP3A family and P-glycoprotein expressed in the GI tract and liver. There is substantial interindividual variation in oral absorption and metabolism of these drugs. Dosing is based on monitoring drug levels.
- Calcineurin inhibitors are associated with a variety of toxicities. Chief among them is nephrotoxicity. As with all immunosuppressants, infections are possible. Gingival hyperplasia and hirsutism are common side effects of cyclosporine.
- Mycophenolate mofetil (MMF) and azathioprine (AZA) block lymphocyte proliferation by inhibiting nucleic acid synthesis. MMF (active mycophenolic acid) is a potent, reversible, noncompetitive inhibitor of IMPDH, which blocks the de novo formation of guanosine monophosphate, blocking T and B cell proliferation. AZA is a prodrug converted to 6-mercaptopurine and then to cytotoxic 6-thioguanine, which is incorporated into nucleic acid chains, blocking further elongation of the DNA. They are used as adjuncts with calcineurin inhibitors with or without a glucocorticoid. MMF has largely replaced AZA due to improved safety and efficacy profile.

- Belatacept is a recombinant fusion protein of CTLA-4. It is a costimulation blocker by binding CD80/86 on APCs, which inhibits T cell activation. It is intended as a substitute for calcineurin inhibitors to avoid the long-term nephrotoxic, cardiovascular, and metabolic complications seen with cyclosporine and tacrolimus. Belatacept increases the risk of PTLD involving the nervous system. Therefore, it is contraindicated in patients who are seronegative to EBV. Serological titers to EBV are typically obtained to confirm exposure.
- The mTOR inhibitors, sirolimus and everolimus, bind FKBP-12. The complex blocks mTOR leading to inhibition of the IL-2 signal transduction pathway blocking it in the G1-S phase, preventing cell cycle progression, which inhibits T cell proliferation. They are frequently used in multidrug regimens to allow reduction of the calcineurin inhibitors. mTOR inhibitors are associated with impaired wound healing/wound dehiscence, leukopenia, thrombocytopenia, anemia, hypercholesterolemia, hypertriglyceridemia, diarrhea, peripheral edema, and other adverse effects. They are substrates of CYP3A4 and P-glycoprotein. They have low bioavailability, extensive hepatic metabolism, and high potential for drug interactions.